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# CATS VS DOGS DATASET ANALYSIS
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import zipfile
import os
import matplotlib.pyplot as plt
from PIL import Image

# Extract ZIP File
zip_file = "data.zip"

with zipfile.ZipFile(zip_file, 'r') as zip_ref:
    zip_ref.extractall("dataset")

print("Dataset Extracted Successfully!")

# Count Images
cat_count = 0
dog_count = 0

cat_images = []
dog_images = []

for root, dirs, files in os.walk("dataset"):
    for file in files:
        if file.lower().endswith(('.jpg', '.jpeg', '.png')):
            full_path = os.path.join(root, file)

            if "cat" in file.lower():
                cat_count += 1
                cat_images.append(full_path)

            elif "dog" in file.lower():
                dog_count += 1
                dog_images.append(full_path)

print("\nNumber of Cat Images :", cat_count)
print("Number of Dog Images :", dog_count)

# Display Sample Cat Image
if len(cat_images) > 0:
    img = Image.open(cat_images[0])

    plt.figure(figsize=(4,4))
    plt.imshow(img)
    plt.axis("off")
    plt.title("Sample Cat Image")
    plt.show()

# Display Sample Dog Image
if len(dog_images) > 0:
    img = Image.open(dog_images[0])

    plt.figure(figsize=(4,4))
    plt.imshow(img)
    plt.axis("off")
    plt.title("Sample Dog Image")

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plt.show()

# Bar Chart
plt.figure(figsize=(6,4))
plt.bar(["Cats", "Dogs"], [cat_count, dog_count])
plt.title("Cats vs Dogs Dataset")
plt.ylabel("Number of Images")
plt.show()

# Pie Chart
plt.figure(figsize=(6,6))
plt.pie(
    [cat_count, dog_count],
    labels=["Cats", "Dogs"],
    autopct='%1.1f%%'
)
plt.title("Cats vs Dogs Distribution")
plt.show()

print("\nProject Completed Successfully!")
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from google.colab import drive
drive.mount('/content/drive')
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Mounted at /content/drive